

Import Libraries

```
import pandas as pd
import numpy as np
```

Loading Dataset

```
data = pd.read_csv("/content/IRIS_Dataset.csv")
```

```
data.shape
```

```
(150, 5)
```

```
data.columns
```

```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
```

```
data.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: [Generate code with data](#) [New interactive sheet](#)

Checking Dataset Information

```
print(data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

```
print(data.isnull().sum)
```

```
<bound method DataFrame.sum of
0      False      False      False      False      False
1      False      False      False      False      False
2      False      False      False      False      False
3      False      False      False      False      False
4      False      False      False      False      False
..      ...      ...      ...      ...      ...
145     False      False      False      False      False
146     False      False      False      False      False
147     False      False      False      False      False
148     False      False      False      False      False
149     False      False      False      False      False
```

```
[150 rows x 5 columns]>
```

Converting Species into Numbers

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()
data['species'] = le.fit_transform(data['species'])
```

Select Features and Target

```
X = data[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']]
y = data['species']
```

Splitting Dataset

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size = 0.2,random_state = 42)
```

Creating Classification Model

```
from sklearn.neighbors import KNeighborsClassifier
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

Training Model

```
model.fit(X_train, y_train)
```

```
▼ KNeighborsClassifier ⓘ ?
KNeighborsClassifier(n_neighbors=3)
```

Predict Species

```
y_pred = model.predict(X_test)
```

Check Accuracy

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 1.0
```

Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Show Species Count

```
species_count = data['species'].value_counts()
print("\nSpecies Count:")
print(species_count)
print("\nExplanation:")
print("0 = Iris-setosa")
print("1 = Iris-versicolor")
print("2 = Iris-virginica")
```

```
Species Count:
species
0      50
1      50
2      50
Name: count, dtype: int64
```

Explanation:
0 = Iris-setosa
1 = Iris-versicolor
2 = Iris-virginica

Creating Graph

```
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
plt.figure(figsize=(3,3))  
sns.countplot(x='species', data=data)  
plt.title("Iris Flower Species Count")  
plt.xlabel("Species")  
plt.ylabel("Count")  
plt.show()
```

